

Enhancing Customer Support with Sentiment Analysis and Language Modeling

Abbott Diabetes Care (ADC) manages customer support across multiple platforms, including its website, social media, chat functionality, phone, and email. In my Module 3 report, clustering analysis was shown to reduce redundancy and improve efficiency by grouping FAQs and transcripts into meaningful categories. Building on that foundation, this report demonstrates how sentiment analysis and large language modeling (LLM) can further enhance ADC's customer support capability. By interpreting emotional tone and intent across channels, ADC can improve responsiveness, protect brand reputation, and strengthen competitiveness in digital health.

Value Chain Activities Essential to Sentiment Analysis

Customer support is the most essential value chain activity for sentiment analysis. Responsiveness occurs not only through email, phone, and chat, but also in social media spaces where customers post questions, concerns, or even trolling content. Sentiment analysis enables ADC to distinguish between genuine issues requiring escalation, FAQs that can be addressed with standardized responses, and disruptive posts that may require moderation. This capability improves satisfaction and safeguards brand trust in public forums.

Language Modeling Approach, Data, and Expected Results

The proposed approach combines clustering with sentiment analysis. Dictionary-based methods provide a baseline, syntactic parsing captures grammar and nuance, and vectorization/LLMs interpret slang, emojis, and context. Advanced architectures such as Mixture of Experts (MoE) (Shazeer et al., 2017; Zhang et al., 2025) and Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) can further improve accuracy and grounding.

- **Data Sources:** FAQs, chat transcripts, call logs, email inquiries, and social media UGC.
- **Platforms/Channels:** Website, social media platforms, phone support notes & call recordings, and customer messaging.

Expected results include detecting sentiment trends within clusters, prioritizing urgent negative cases, and providing agents and social media managers with context-aware response suggestions.

Resources, Risks & Mitigation

ADC requires cloud NLP services, CRM/ticketing integration, and secure data pipelines supported by cross-functional collaboration across Marketing, Product, IT, Customer Support, Regulatory, and UX is essential. Compliance oversight ensures HIPAA and FDA standards are met. Human oversight is needed to validate AI-generated sentiment classifications, especially in public social media contexts. Risks include misclassification of sarcasm or slang, reliance on cloud partners, and LLM hallucinations. Mitigation strategies include domain-specific retraining, human-in-the-loop validation, hybrid deployment, and retrieval-grounded architectures.

Expected Enhancement

By extending clustering analysis with sentiment detection and large language modeling, ADC can move from simply organizing customer inquiries to actively interpreting their emotional tone and urgency. This enhancement allows support teams to identify clusters of negative sentiment early, prioritize high-risk issues such as product malfunctions or insurance frustrations, and deliver more empathetic responses. In social media contexts, sentiment analysis helps managers choose the right response strategy — whether to escalate, provide a standardized FAQ link, or moderate disruptive content. The integration of sentiment analysis into clustered data provides a competitive advantage by reducing agent triage time, improving customer satisfaction scores, and enabling proactive communication strategies.

Conclusion

By extending clustering analysis with sentiment detection and language modeling, ADC can transform customer support into a proactive, insight-driven capability. This approach enhances competitiveness by improving responsiveness across all channels — email, phone, chat, and especially social media. It reduces churn, strengthens trust, and protects brand reputation in public forums. While risks such as misclassification and reliance on cloud partners exist, they can be mitigated through retraining, human oversight, and retrieval-grounded architectures. This demonstrates how unsupervised learning and advanced NLP can be embedded into ADC's value chain to deliver measurable business impact.

APPENDIX:

Table 1: Clustering vs. Clustering + Sentiment Analysis Comparison

Dimension	Clustering (Module 3)	Clustering + Sentiment/LLM (Module 5)
Primary Goal	Organize FAQs, chat logs, and call transcripts into meaningful groups	Interpret both <i>topics</i> and <i>emotional tone</i> of inquiries
Data Inputs	Text content (FAQs, transcripts, queries) + metadata (frequency, channel)	Same inputs plus UGC (social media posts, reviews, emojis)
Analytical Method	Unsupervised machine learning (tokenization, TF-IDF, embeddings)	Sentiment analysis (dictionary, syntactic parsing, vectorization) + LLMs (MoE, RAG)
Output	Clusters of related inquiries (e.g., “sensor adhesion,” “app connectivity”)	Clusters enriched with sentiment scores (positive, negative, neutral, emotions)
Business Value	Reduced redundancy, streamlined regulatory review, improved navigation	Faster triage of urgent complaints, proactive detection of issues, empathetic agent responses
Risks	Clusters may reflect noise or bias if not validated	Misclassification of slang/sarcasm, reliance on cloud NLP, LLM hallucinations
Mitigation	Cross-validation, pruning, regulatory oversight	Domain-specific retraining, human-in-the-loop validation, retrieval grounding (RAG)

References

- Clark, J. R. (2025). *Improving customer support capability at Abbott Diabetes Care using cluster analysis* (Module 3 Report, MIS6250: Strategic AI for Business). Northeastern University.
- Lewis, P., Perez, E., Piktus, A., Karpukhin, V., Goyal, N., Kulikov, M., ... & Riedel, S. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems*, 33, 9459–9474.
https://papers.nips.cc/paper_files/paper/2020/file/6b493230205f780e1bc26945df7481e5-Paper.pdf
- Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., & Dean, J. (2017). Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. *arXiv preprint arXiv:1701.06538*. <https://arxiv.org/abs/1701.06538>
- Zhang, D., Song, J., Bi, Z., Yuan, Y., Wang, T., Yeong, J., & Hao, J. (2025). Mixture of experts in large language models. *arXiv preprint arXiv:2507.11181*. <https://arxiv.org/abs/2507.11181>